

Multiple Choice (10 marks)

1. In Python 3, what is the output of `['Hi!'] * 4`?

`'Hi!', 'Hi!', 'Hi!', 'Hi!'`

`'Hi!' * 4`

- (a) Error
 - (b) None of the above.
2. What is the output of `"abc"[-1]` in Python 3?
- (a) Error
 - (b) a
 - (c) b
 - (d) c
3. Which of the following best describes the primary way in which a distributed denial-ofservice (D D o S) attack differs from a denial-ofservice (D o S) attack?
- (a) The goal of the attack
 - (b) The number of computers being attacked
 - (c) The number of computers launching the attack
 - (d) The time period in which the attack occurs
4. Which of the following is the hexadecimal representation of the decimal number 231_{10} ?
- (a) 17_{16}
 - (b) $E4_{16}$
 - (c) $E7_{16}$
 - (d) $F4_{16}$
5. Let `l = [1,2,2,3,4]`. In Python 3, what is a possible output of `set(l)`?
- (a) `{1}`
 - (b) `{1,2,2,3,4}`
 - (c) `{1,2,3,4}`
 - (d) `{4,3,2,2,1}`

True / False (10 marks)

Answer True or False

6. True or False: Authentication is compromised when someone uses stolen login credentials to access a system.
7. True or False: The function $O(N^{1/N})$ grows the slowest among the given types of functions as N increases.
8. True or False: The boolean expression $a[i] == \max \parallel !(\max != a[i])$ simplifies to $a[i] < \max \parallel a[i] > \max$.
9. True or False: In Python 3, floor division is represented by the operator `'/'`.
10. True or False: The method `'sort()'` can be used directly on string objects to sort their characters in Python.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the importance of clearly defining problem specifications in programming. How should a method be designed to handle cases with multiple values larger than a given item in an array?
12. Discuss the potential benefits of an upgraded system in a company. In your analysis, identify which benefit is least likely to be realized and explain your reasoning.

End of Paper